

NAME

email@gmail.com | <https://www.linkedin.com> | <https://github.com>

EDUCATION

The Ohio State University

B.S. in Computer Science & Engineering

Columbus, Ohio

Expected Graduation May 2028

WORK EXPERIENCE

ReRoute Lab

Columbus, Ohio

Undergraduate Research

May 2025 - Present

- Developed an internal data-processing pipeline in Python to **process and visualize 10,000+ wildlife images** used in research reports and decision-making for ecologists studying African wild dogs
- Automated reporting by merging drone and camera data into CSV tables, cutting manual preparation time by **~70%**.
- Produced an AI pipeline integrating bioacoustic recordings, visitor photos, drone footage, and camera trap data, improving the Wilds' understanding of species behavior, influencing conservation decisions for Père David's deer
- Used multi-modal data to merge audio using **BirdNET** with visual detections using **YOLO** for richer ecological insights

Accenture

Columbus, Ohio

SWE Intern

March 2024 - May 2024

- Created an interactive **Java-based Tomodachi pet application**, applying object-oriented design principles
- Worked in a **3-member Agile Scrum team**, participating in sprint planning and daily stand-ups
- Used **Git-based version control** for development, including branches, pull requests, and merge conflict resolution
- Gained hands-on experience with **CI/CD pipeline concepts** aligned with Accenture's development practices

PROJECTS

Backend Email Newsletter

Present

- Developed a REST API in **Rust** using **actix-web** and **Tokio**, with health monitoring and subscriber endpoints
- Designed **unit and integration tests** to validate API behavior and input constraints, improving reliability
- Integrated **PostgreSQL** persistence with **SQLx**, using connection pooling and error handling for database interactions
- Containerized the application using **multi-stage Docker builds**, optimizing image size and performance

Custom Unix Shell

Dec 2026 - Jan 2026

- Implemented a Unix-like shell using the Rust standard library, featuring a full REPL, environment variable handling, parent-child process coordination, and execution of external programs via OS process spawning
- Designed a command parser handling quoting rules, argument tokenization, pipelines, and stdout/stderr redirection
- Built persistent command history with interactive navigation and file-backed storage across shell sessions
- Tested edge cases such as redirection ordering, child process lifecycle management, and filesystem state mutations

INVOLVEMENT

Choose Ohio First (Scholarship Program)

Aug 2024 - Present

- Completed **40+ volunteer hours per semester** through community-focused STEM outreach initiatives
- Participated in the Wonders of the World program, leading science experiments for students in Columbus City Schools

Big Data Analytics Program

Aug 2024 - Present

- Active member of OSU's data analytics organization focused on applied data science and industry-relevant projects
- Earned **3rd in Data IO 2024** by completing an EV charging infrastructure optimization analysis, processing U.S. EV charger and vehicle registration datasets using **Pandas**, and generating heat maps using **AWS Quicksight**

PUBLICATION

- Kline, J., Potlapally, A., Pillai, B., Wani, T., Katole, R., Patil, V., Covey, P., Subramoni, H., Berger-Wolf, T., & Stewart, C. (2025). *SmartWilds: Multimodal Wildlife Monitoring Dataset*. **Imageomics Workshop, NeurIPS 2025**. arXiv:2509.18894. <https://doi.org/10.48550/arXiv.2509.18894>
- Multimodal HeatMaps Wilds: Multimodal Wildlife Monitoring Dataset Under submission — **CVPR** (Computer Vision and Pattern Recognition)

SKILLS

Languages: Java, Python, Rust, MATLAB, PostgreSQL, Docker, Linux, GitHub, PyTorch, Pandas, NumPy, YOLO, AWS, GCP

Certifications: AWS Certified Cloud Practitioner, Google Cloud Digital Leader, Batch System on OSC

Relevant Coursework: Data Structures and Algorithms, Systems 1, Software 2, Intermediate Studies in AI, Electronics for CS